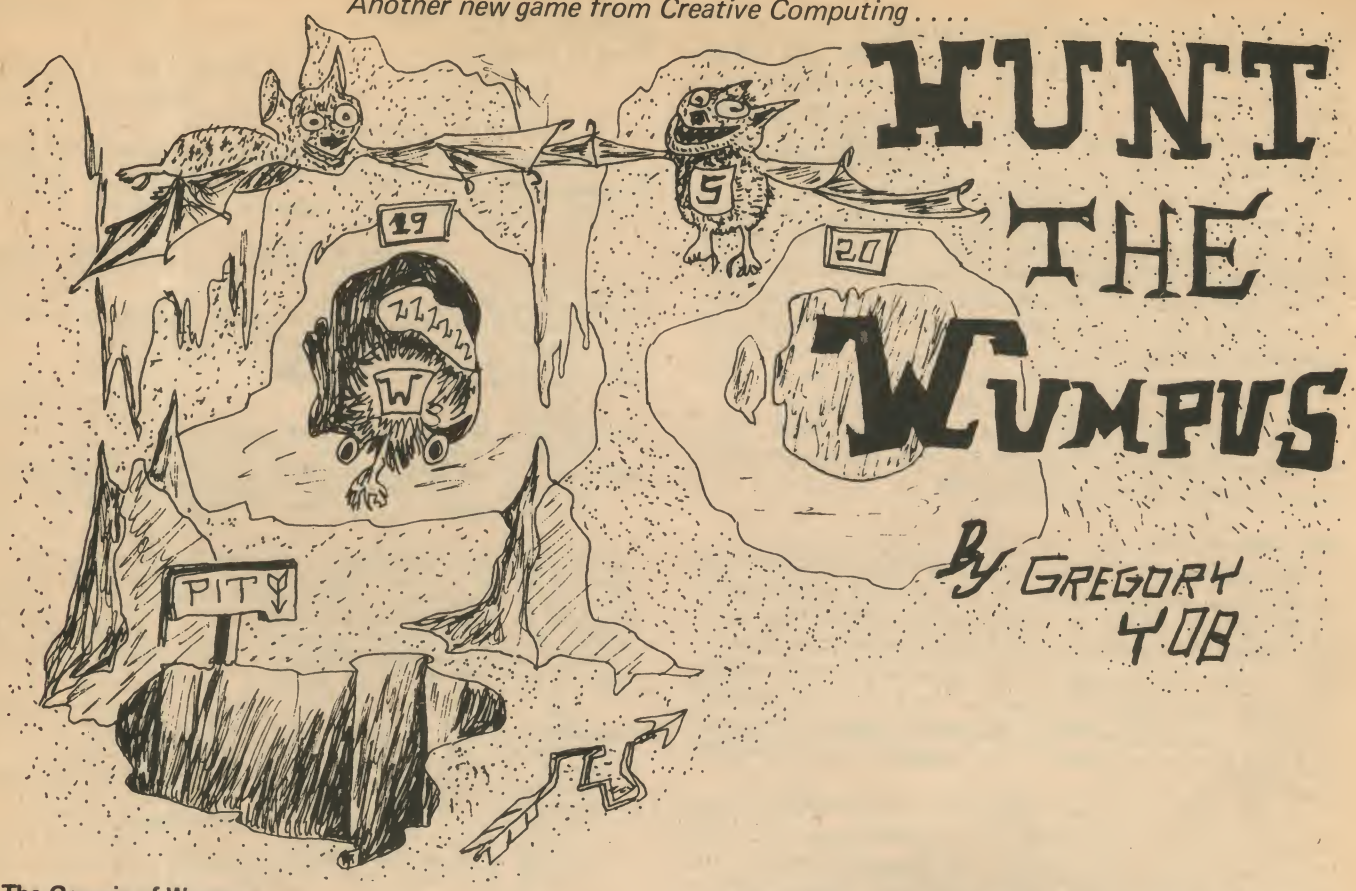
Another new game from Creative Computing . . . .



HUNT THE WUMPUS

By GREGORY YOB

The Genesis of Wumpus

Two years ago I happened by People's Computer Company (PCC) and saw some of their computer games — such as Hurkle, Snark, and Mugwump. My reaction was: "EECH!!" Each of these games was based on a 10 x 10 grid in Cartesian co-ordinates and three of them was too much for me. I started to think along the lines of: "There has to be a hide and seek computer game without that (exp. deleted) grid!!" In fact, why not a topological computer game — Imagine a set of points connected in some way and the player moves about the set via the interconnections.

That afternoon in meditation the phrase "Hunt the Wumpus" arrived, and Wumpus was born. He's still a bit vague in physical detail as most dedicated Wumpus hunters know, but appearances are part of the game. (If you like, send me a picture of your version of a Wumpus. Perhaps friendly Dave, our editor, will publish the best one in *Creative Computing*.) The grid I chose was the vertices of a dodecahedron — simply because it's my favorite Platonic solid and once, ages ago, I made a kite shaped like one. The edges became the connecting tunnels between the caves which were the set of points for the game.

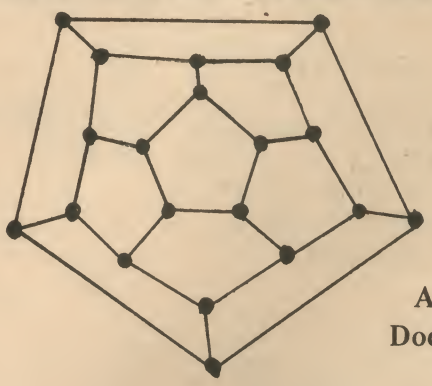
My basic idea at this time was for the player to approach the Wumpus, back off, and come up to him by going around the dodecahedron. To my knowledge, this has never happened . . . most players adopt other strategies rather than this cold-blooded approach.

Anyway . . . how to get the Wumpus! How about an arrow which could turn corners as it goes from room to room. Let the hunter tell the arrow where to go and let it fly. The shortest round trip without reversals is 5 caves — and thus the Crooked Arrow.

Hmmm . . . How does one sense the Wumpus? It's dark in yonder cave, and light would wake him up. If one got one cave away, the wumpus's distinct smell would serve as a warning. So far, so good . . . but Wumpus is still too easy, so let's find some appropriate hazards for the caves.

Bottomless pits were easy. Any imaginary cave would have a few of those around the place. Superbats were harder to come by. It took me a day or two to get that idea. The Superbats are a sort of rapid transit system gone a little batty (sorry about that one). They take you a random distance to a random cave and leave you there. If that's a pit or a Wumpus, well, you are in Fate's hands.

Around this time, I saw that Map-making would be a regular activity of Wumpus-hunters. I numbered the caves and made the scheme fixed in the hopes a practised player might notice this and make himself a permanent map of the caverns. (Another unrealised hope — as an exercise, make yourself such a map on a Squashed Dodecahedron).



A Squashed Dodecahedron

To start the game fairly, Wumpus, Hazards, and Hunter are located on different points at the start of the game. Each game starts with random choices of location, but the hunter may restart with the same set-up if he chooses. This allows re-plays if the hunter, say, fell into a pit on the first move.

Wumpus was nearly done in my mind . . . (hint to a games-writer: Have a clear notion of your game before you

start coding it. This saves MUCH confusion.) yet I felt it was a bit dull. Once you found the Wumpus all you had to do was shoot it. To fix this, the Wumpus was given a little life. If you shot an arrow or moved into his cave, he woke up and chose to move to a neighboring room or to the same room (one of 4 choices). If you and the Wumpus were in the same room after he moved, he ATE YOU UP!!

Around here I noticed that the pits and the bats didn't affect the Wumpus. To explain this, I added some color by making him heavy and with the legendary sucker feet. After all, evolution works in strange ways!! If you are a Wumpus fiend, make a version of Wumpus in which he avoids pits and superbats can carry him only one room (with the possibility of being dumped into your cave). This can be done by making the wumpus moving procedure a subroutine.

I wrote Wumpus and dropped it off at PCC. Then I went home and dreamed up Wumpus II which will be covered in the next issue of *Creative Computing*.

The Birth of Wumpus

Around a month later, I went to the Synergy conference at Stanford, where many of the far-out folk were gathered to share their visions of improving the world. PCC had a few terminals running in a conference room and I dropped by. To my vast surprise, all of the terminals were running Wumpus and scraps of paper on the floor with scrawled numbers and lines testified that much dedicated Wumpus-hunting was in progress. I had spawned a hit computer game!!!

Later, PCC published Wumpus in its newsletter (If you haven't seen it, write them for a subscription: P.O. Box 310, Menlo Park, Cal. 94025), and Wumpus appeared in all sorts of unlikely places. I have reports of Wumpus written in RPG, a listing of one in FORTRAN, a rumor of a system command of 'to Wumpus' on a large corporation's R&D computer system and have even seen an illustrated version for the Hazeltine CRT terminal!!

WUMPUS TAPES, ETC.

I can be found at:

Gregory Yob
PO Box 354
Palo Alto, Calif. 94301

Paper tapes of Wumpus, Wumpus 2 and Wumpus 3 are available and cost $5.00 each.

May your arrows remain straight. —Gregory Yob.

SAMPLE RUN

```
INSTRUCTIONS (Y-N)?Y
WELCOME TO 'HUNT THE WUMPUS'
  THE WUMPUS LIVES IN A CAVE OF 20 ROOMS. EACH ROOM
HAS 3 TUNNELS LEADING TO OTHER ROOMS. (LOOK AT A
DODECAHEDRON TO SEE HOW THIS WORKS-IF YOU DON'T KNOW
WHAT A DODECAHEDRON IS, ASK SOMEONE)

    HAZARDS:
 BOTTOMLESS PITS - TWO ROOMS HAVE BOTTOMLESS PITS IN THEM
     IF YOU GO THERE, YOU FALL INTO THE PIT (& LOSE!)
 SUPER BATS - TWO OTHER ROOMS HAVE SUPER BATS. IF YOU
     GO THERE, A BAT GRABS YOU AND TAKES YOU TO SOME OTHER
     ROOM AT RANDOM. (WHICH MIGHT BE TROUBLESOME)

     WUMPUS:
 THE WUMPUS IS NOT BOTHERED BY THE HAZARDS (HE HAS SUCKER
 FEET AND IS TOO BIG FOR A BAT TO LIFT).  USUALLY
 HE IS ASLEEP. TWO THINGS WAKE HIM UP: YOUR ENTERING
 HIS ROOM OR YOUR SHOOTING AN ARROW.
     IF THE WUMPUS WAKES, HE MOVES (P=.75) ONE ROOM
 OR STAYS STILL (P=.25). AFTER THAT, IF HE IS WHERE YOU
 ARE, HE EATS YOU UP (& YOU LOSE!)

     YOU:
 EACH TURN YOU MAY MOVE OR SHOOT A CROOKED ARROW
   MOVING: YOU CAN GO ONE ROOM (THRU ONE TUNNEL)
   ARROWS: YOU HAVE 5 ARROWS. YOU LOSE WHEN YOU RUN OUT.
   EACH ARROW CAN GO FROM 1 TO 5 ROOMS. YOU AIM BY TELLING
   THE COMPUTER THE ROOM#S YOU WANT THE ARROW TO GO TO.
   IF THE ARROW CAN'T GO THAT WAY(IE NO TUNNEL) IT MOVES
   AT RANDOM TO THE NEXT ROOM.
     IF THE ARROW HITS THE WUMPUS, YOU WIN.
     IF THE ARROW HITS YOU, YOU LOSE.

    WARNINGS:
     WHEN YOU ARE ONE ROOM AWAY FROM WUMPUS OR HAZARD,
    THE COMPUTER SAYS:
 WUMPUS-  'I SMELL A WUMPUS'
 BAT   -  'BATS NEARBY'
 PIT   -  'I FEEL A DRAFT'
```

```
HUNT THE WUMPUS


BATS NEARBY!
YOU ARE IN ROOM  2
TUNNELS LEAD TO  1      3      10

SHOOT OR MOVE (S-M)?M
WHERE TO?1
ZAP--SUPER BAT SNATCH! ELSEWHEREVILLE FOR YOU!
YYYIIIIFFFF . . . FELL IN PIT
HA HA HA - YOU LOSE!
SAME SET-UP (Y-N)?Y
HUNT THE WUMPUS


BATS NEARBY!
YOU ARE IN ROOM  2
TUNNELS LEAD TO  1      3      10

SHOOT OR MOVE (S-M)?M
WHERE TO?3

YOU ARE IN ROOM  3
TUNNELS LEAD TO  2      4      12
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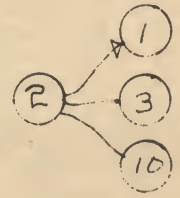
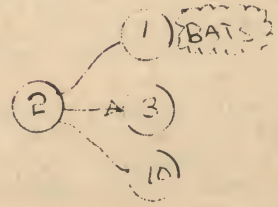




SUPER BA...
PUT ME ... A
PIT SOMEWHERE

more

```
SHOOT OR MOVE (S-M)?M
WHERE TO?4

YOU ARE IN ROOM  4
TUNNELS LEAD TO  3      5      14

SHOOT OR MOVE (S-M)?M
WHERE TO?5

BATS NEARBY!
YOU ARE IN ROOM  5
TUNNELS LEAD TO  1      4      6

SHOOT OR MOVE (S-M)?M
WHERE TO?6

I FEEL A DRAFT
YOU ARE IN ROOM  6
TUNNELS LEAD TO  5      7      15

SHOOT OR MOVE (S-M)?M
WHERE TO?7
YYYIIIIEEEE . . . FELL IN PIT
HA HA HA - YOU LOSE!
SAME SET-UP (Y-N)?Y
HUNT THE WUMPUS

BATS NEARBY!
YOU ARE IN ROOM  2
TUNNELS LEAD TO  1      3      10

SHOOT OR MOVE (S-M)?M
WHERE TO?10

BATS NEARBY!
YOU ARE IN ROOM  10
TUNNELS LEAD TO  2      9      11

SHOOT OR MOVE (S-M)?M
WHERE TO?11
ZAP--SUPER BAT SNATCH! ELSEWHEREVILLE FOR YOU!

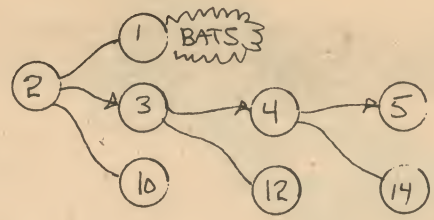
YOU ARE IN ROOM  14
TUNNELS LEAD TO  4      13     15

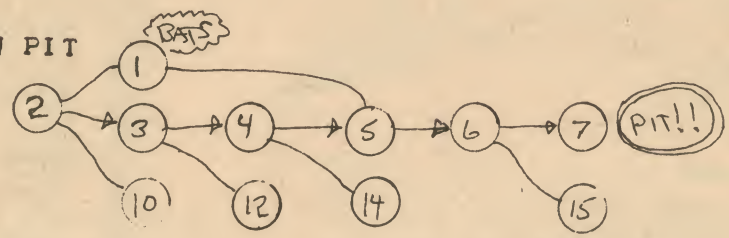
SHOOT OR MOVE (S-M)?M
WHERE TO?15

I SMELL A WUMPUS!
YOU ARE IN ROOM  15
TUNNELS LEAD TO  6      14     16

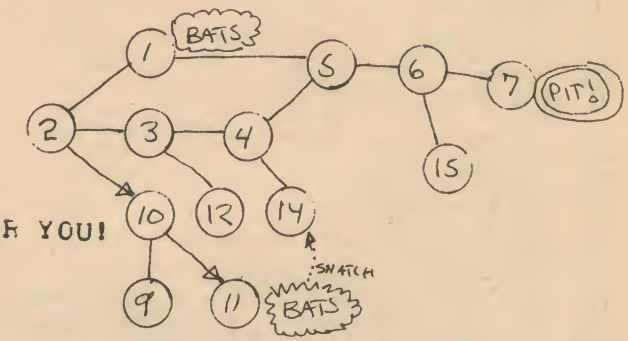
SHOOT OR MOVE (S-M)?S
NO. OF ROOMS(1-5)?1
ROOM #?16
AHA! YOU GOT THE WUMPUS!
HEE HEE HEE - THE WUMPUS'LL GETCHA NEXT TIME!!
```

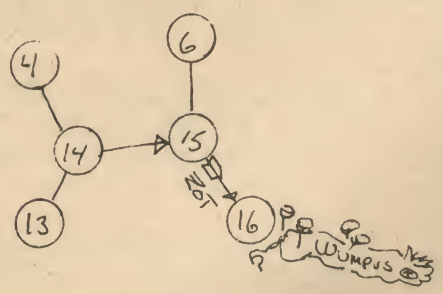
JUST KEEP ON TRUCKIN'!





BLEW IT AGAIN!!





CAN YOU FIT THIS MAP INTO THE OTHER ONE ABOVE? FIGURE OUT HOW I KNEW THE WUMPUS WAS IN 16.

PROGRAM LISTING

```
0010  REM- HUNT THE WUMPUS
0015  REM:  BY GREGORY YOB
0020  PRINT "INSTRUCTIONS (Y-N)";
0030  INPUT I$
0040  IF I$="N" THEN 52
0050  GOSUB 1000
0052  REM- ANNOUNCE WUMPUSII FOR ALL AFICIONADOS ... ADDED BY DAVE
0054  PRINT
0056  PRINT "     ATTENTION ALL WUMPUS LOVERS!!!"
0058  PRINT "     THERE ARE NOW TWO ADDITIONS TO THE WUMPUS FAMILY";
0060  PRINT " OF PROGRAMS."
0062  PRINT
0064  PRINT "     WUMP2:  SOME DIFFERENT CAVE ARRANGEMENTS"
0066  PRINT "     WUMP3:  DIFFERENT HAZARDS"
0067  PRINT
0068  REM- SET UP CAVE (DODECAHEDRAL NODE LIST)
0070  DIM S(20,3)
0080   FOR J=1 TO 20
0090    FOR K=1 TO 3
0100    READ S(J,K)
0110    NEXT K
0120   NEXT J
0130  DATA 2,5,8,1,3,10,2,4,12,3,5,14,1,4,6
0140  DATA 5,7,15,6,8,17,1,7,9,8,10,18,2,9,11
0150  DATA 10,12,19,3,11,13,12,14,20,4,13,15,6,14,16
0160  DATA 15,17,20,7,16,18,9,17,19,11,18,20,13,16,19
0170  DEF FNA(X)=INT(20*RND(0))+1
0180  DEF FNB(X)=INT(3*RND(0))+1
0190  DEF FNC(X)=INT(4*RND(0))+1
0200  REM-LOCATE L ARRAY ITEMS
0210  REM-1-YOU,2-WUMPUS,3&4-PITS,5&6-BATS
0220  DIM L(6)
0230  DIM M(6)
0240   FOR J=1 TO 6
0250   L(J)=FNA(0)
0260   M(J)=L(J)
0270   NEXT J
0280  REM-CHECK FOR CROSSOVERS (IE L(1)=L(2),ETC)
0290   FOR J=1 TO 6
0300    FOR K=J TO 6
0310    IF J=K THEN 330
0320    IF L(J)=L(K) THEN 240
0330    NEXT K
0340   NEXT J
0350  REM-SET# ARROWS
0360  A=5
0365  L=L(1)
0370  REM-RUN THE GAME
0375  PRINT "HUNT THE WUMPUS"
0380  REM-HAZARD WARNINGS & LOCATION
0390  GOSUB 2000
0400  REM-MOVE OR SHOOT
0410  GOSUB 2500
0420  GOTO 0 OF 440,480
0430  REM-SHOOT
0440  GOSUB 3000
0450  IF F=0 THEN 390
0460  GOTO 500
0470  REM-MOVE
0480  GOSUB 4000
0490  IF F=0 THEN 390
0500  IF F>0 THEN 550
0510  REM-LOSE
0520  PRINT "HA HA HA - YOU LOSE!"
0530  GOTO 560
0540  REM-WIN
0550  PRINT "HEE HEE HEE - THE WUMPUS'LL GETCHA NEXT TIME!!"
0560   FOR J=1 TO 6
0570   L(J)=M(J)
0580   NEXT J
0590  PRINT "SAME SET-UP (Y-N)";
0600  INPUT I$
0610  IF I$#"Y" THEN 240
0620  GOTO 360
1000  REM-INSTRUCTIONS
1010  PRINT "WELCOME TO 'HUNT THE WUMPUS'"
1020  PRINT "  THE WUMPUS LIVES IN A CAVE OF 20 ROOMS. EACH ROOM"
1030  PRINT "HAS 3 TUNNELS LEADING TO OTHER ROOMS. (LOOK AT A"
1040  PRINT "DODECAHEDRON TO SEE HOW THIS WORKS-IF YOU DON'T KNOW"
1050  PRINT "WHAT A DODECAHEDRON IS, ASK SOMEONE)"
1060  PRINT
1070  PRINT "     HAZARDS:"
1080  PRINT " BOTTOMLESS PITS - TWO ROOMS HAVE BOTTOMLESS PITS IN THEM"
1090  PRINT "     IF YOU GO THERE, YOU FALL INTO THE PIT (& LOSE!)"
1100  PRINT " SUPER BATS - TWO OTHER ROOMS HAVE SUPER BATS. IF YOU"
1110  PRINT "     GO THERE, A BAT GRABS YOU AND TAKES YOU TO SOME OTHER"
1120  PRINT "     ROOM AT RANDOM. (WHICH MIGHT BE TROUBLESOME)"
1130  PRINT
1140  PRINT "     WUMPUS:"
1150  PRINT " THE WUMPUS IS NOT BOTHERED BY THE HAZARDS (HE HAS SUCKER"
1160  PRINT " FEET AND IS TOO BIG FOR A BAT TO LIFT).  USUALLY"
1170  PRINT " HE IS ASLEEP. TWO THINGS WAKE HIM UP: YOUR ENTERING"
1180  PRINT " HIS ROOM OR YOUR SHOOTING AN ARROW."
1190  PRINT "     IF THE WUMPUS WAKES, HE MOVES (P=.75) ONE ROOM"
1200  PRINT " OR STAYS STILL (P=.25). AFTER THAT, IF HE IS WHERE YOU"
1210  PRINT " ARE, HE EATS YOU UP (& YOU LOSE!)"
1220  PRINT
1230  PRINT "     YOU:"
1240  PRINT " EACH TURN YOU MAY MOVE OR SHOOT A CROOKED ARROW"
1250  PRINT "   MOVING: YOU CAN GO ONE ROOM (THRU ONE TUNNEL)"
1260  PRINT "   ARROWS: YOU HAVE 5 ARROWS. YOU LOSE WHEN YOU RUN OUT."
1270  PRINT "   EACH ARROW CAN GO FROM 1 TO 5 ROOMS. YOU AIM BY TELLING"
1280  PRINT "   THE COMPUTER THE ROOM#S YOU WANT THE ARROW TO GO TO."
1290  PRINT "   IF THE ARROG CAN'T GO DHAT WAY(IE NO TUNNEL) IT MOFES"
1300  PRINT "   AT RANDOM TO THE NEXT ROOM."
1310  PRINT "     IF THE ARROW HITS THE WUMPUS, YOU WIN."
1320  PRINT "     IF THE ARROW HITS YOU, YOU LOSE."
1330  PRINT
1340  PRINT "    WARNINGS:"
1350  PRINT "     WHEN YOU ARE ONE ROOM AWAY FROM WUMPUS OR HAZARD,"
1360  PRINT "    THE COMPUTER SAYS:"
1370  PRINT " WUMPUS-  'I SMELL A WUMPUS'"
1380  PRINT " BAT   -  'BATS NEARBY'"
1390  PRINT " PIT   -  'I FEEL A DRAFT'"
1400  PRINT ""
1410  RETURN
2000  REM-PRINT LOCATION & HAZARD WARNINGS
2010  PRINT
2020   FOR J=2 TO 6
2030    FOR K=1 TO 3
2040    IF S(L(1),K)#L(J) THEN 2110
2050    GOTO J-1 OF 2060,2080,2080,2100,2100
2060    PRINT "I SMELL A WUMPUS!"
2070    GOTO 2110
2080    PRINT "I FEEL A DRAFT"
2090    GOTO 2110
2100    PRINT "BATS NEARBY!"
2110    NEXT K
2120   NEXT J
2130  PRINT "YOU ARE IN ROOM "L(1)
2140  PRINT "TUNNELS LEAD TO "S(L,1);S(L,2);S(L,3)
2150  PRINT
2160  RETURN
2500  REM-CHOOSE OPTION
2510  PRINT "SHOOT OR MOVE (S-M)";
2520  INPUT I$
2530  IF I$#"S" THEN 2560
2540  O=1
2550  RETURN
2560  IF I$#"M" THEN 2510
2570  O=2
2580  RETURN
3000  REM-ARROW ROUTINE
3010  F=0
3020  REM-PATH OF ARROW
3030  DIM P(5)
3040  PRINT "NO. OF ROOMS(1-5)";
3050  INPUT J9
3060  IF J9<1 OR J9>5 THEN 3040
3070   FOR K=1 TO J9
3080   PRINT "ROOM #";
3090   INPUT P(K)
3095   IF K <= 2 THEN 3115
3100   IF P(K) <> P(K-2) THEN 3115
3105   PRINT "ARROWS AREN'T THAT CROOKED - TRY ANOTHER ROOM"
3110   GOTO 3080
3115   NEXT K
3120  REM-SHOOT ARROW
3130  L=L(1)
3140   FOR K=1 TO J9
3150    FOR K1=1 TO 3
3160    IF S(L,K1)=P(K) THEN 3295
3170    NEXT K1
3180   REM-NO TUNNEL FOR ARROW
3190   L=S(L,FNB(1))
3200   GOTO 3300
3210   NEXT K
3220  PRINT "MISSED"
3225  L=L(1)
3230  REM-MOVE WUMPUS
3240  GOSUB 3370
3250  REM-AMMO CHECK
3255  A=A-1
3260  IF A>0 THEN 3280
3270  F=-1
3280  RETURN
3290  REM-SEE IF ARROW IS AT L(1) OR L(2)
3295  L=P(K)
3300  IF L#L(2) THEN 3340
3310  PRINT "AHA! YOU GOT THE WUMPUS!"
3320  F=1
3330  RETURN
3340  IF L#L(1) THEN 3210
3350  PRINT "OUCH! ARROW GOT YOU!"
3360  GOTO 3270
3370  REM-MOVE WUMPUS ROUTINE
3380  K=FNC(0)
3390  IF K=4 THEN 3410
3400  L(2)=S(L(2),K)
3410  IF L(2)#L THEN 3440
3420  PRINT "TSK TSK TSK- WUMPUS GOT YOU!"
3430  F=-1
3440  RETURN
4000  REM- MOVE ROUTINE
4010  F=0
4020  PRINT "WHERE TO";
4030  INPUT L
4040  IF L<1 OR L>20 THEN 4020
4050   FOR K=1 TO 3
4060   REM- CHECK IF LEGAL MOVE
4070   IF S(L(1),K)=L THEN 4130
4080   NEXT K
4090  IF L=L(1) THEN 4130
4100  PRINT "NOT POSSIBLE -";
4110  GOTO 4020
4120  REM-CHECK FOR HAZARDS
4130  L(1)=L
4140  REM-WUMPUS
4150  IF L#L(2) THEN 4220
4160  PRINT "... OOPS! BUMPED A WUMPUS!"
4170  REM-MOVE WUMPUS
4180  GOSUB 3380
4190  IF F=0 THEN 4220
4200  RETURN
4210  REM-PIT
4220  IF L#L(3) AND L#L(4) THEN 4270
4230  PRINT "YYYIIIIEEEE . . . FELL IN PIT"
4240  F=-1
4250  RETURN
4260  REM-BATS
4270  IF L#L(5) AND L#L(6) THEN 4310
4280  PRINT "ZAP--SUPER BAT SNATCH! ELSEWHEREVILLE FOR YOU!"
4290  L=FNA(1)
4300  GOTO 4130
4310  RETURN
5000  END
```